2. **Specific Gravity:** Specific gravity is the ratio of the density of a substance to the density of a reference substance (usually water). Specific gravity of petrol should be 0.70 to 0.78
3. **Calorific Value:** The amount of heat energy produced by the complete combustion when burning 1 kg of a fuel. The calorific value of petrol is 45.8 MJ/kg.
4. **Flash and Fire Point:** The fire point of a fuel is the lowest temperature at which the vapor of that fuel will continue to burn for at least 5 seconds after ignition by an open flame. Generally flash point should be high for the fuel. The fire point of a fuel is the temperature at which the vapor produced by that given fuel will continue to burn for at least 5 seconds after ignition by an open flame. In general the fire points can be assumed to be about 15°C - 20°C higher than the flash points. For petrol, at least 10% of fuel should be burn instantly and rest in staged phase.
5. **Viscosity:** Resistance to the flow is called as Viscosity and it should be low.
6. **Sulphur Content:** Sulphur will corrode and damage the metal parts. During engine operation, Sulphur combines with oxygen to form Sulphur-di-oxide and in presence of water, it forms sulphurous acid. Hence, Sulphur content should be less than 0.1%
7. **Moisture and Sediment Content:** Petrol fuel should be free from Moisture and Sediment Content.
8. **Octane Number:** This is a measure of auto Ignition resistance in a spark-ignition engine. It represents the volume percentage of iso-octane (C_8H_{18}) in a iso-octane (C_8H_{18}) / n-heptane (C_7H_{16}) mixture. Higher the rating, higher the resistance to knock.A higher rating does not indicate more power but fuel can be used in higher compression ratio. The value of Octane number for the available fuel is between 85 to 90.

3.1.2.2 Diesel and Its Properties

Diesel fuel is the light oil and is obtain from crude oil by the distillation process at a temperature of 250°C – 300°C. Diesel consists of 85% carbon, 12% Hydrogen and 3 % others by weight. These have a boiling point between 250°C and 350°C. Diesel contains more energy than petrol. A diesel engine can be up to 40% more efficient than a spark-ignited petrol engine with the same power output and hence it is widely used in cars, trucks, buses, railway engines etc., The following are the required properties of diesel.

1. **Volatility:** The volatility of diesel is less than petrol. The volatility of diesel fuel influences density, auto ignition temperature, flash point, viscosity and cetane number. High volatility promotes vapor lock and low volatility component may not burn completely, thereby increasing smoke deposits.
2. **Specific Gravity:** Specific gravity of diesel is higher than petrol and the value should be 0.82 to 0.92
3. **Calorific Value:** The amount of heat energy produced by the complete combustion when burning 1 kg of a fuel. The calorific value of diesel is slightly lesser than petrol and the value is 45.5 MJ/kg
4. **Viscosity:** The viscosity is a measure of the resistance to flow of the fuel.

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